Activity 10 Pre-Lab — Crossed Aldol (2-Acetylpyridine + 4-Nitrobenzaldehyde)

[Your name] & [partner]
[MM/DD/YY]

Organic Lab · aldol addition vs. condensation

■ **Black = write this in your notebook** ■ **Blue = context, don't copy it** Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Run a base-catalyzed **crossed aldol** of **2-acetylpyridine** (the enolate) with **4-nitrobenzaldehyde** (the electrophile), and show how **reaction conditions** decide the product: **room temp** → **aldol addition** (β-hydroxy ketone) vs **heat (50 °C)** → **aldol condensation** (the α,β-unsaturated ketone + H_2O).

✍ **DRAW — both outcomes**
input chemical structures: 2-acetylpyridine + 4-nitrobenzaldehyde
→ **addition product** (β-hydroxy ketone) RT · → **condensation product** (α,β-unsaturated ketone) + H_2O 50 °C, dehydration
base = Na_2CO_3 makes the enolate at the acetyl CH_3; it attacks the aldehyde C=O.

Chemical Reagents & Safety Table

CONTEXT › Required: 2-acetylpyridine, 4-nitrobenzaldehyde, sodium carbonate, methanol — plus a row for each product (MW only, per directions). Typical SDS values — cross-check the SDS tile.

Chemical	Role	MW	Amount	Physical properties	Hazards (GHS)
2-Acetylpyridine	Enolate (nucleophile)	121.14	~0.2 mL (weigh exact)	Yellow liquid; bp 192 °C; d 1.08 g/mL	H302 harmful if swallowed · H315 · H319 · H335
4-Nitrobenzaldehyde	Aldehyde (electrophile)	151.12	300 mg	Pale-yellow solid; mp 106 °C	H315 skin irritation · H319 eye irritation · H335 respiratory irritation
Sodium carbonate (0.54% w/v)	Base (makes enolate)	105.99	10 mL	Colorless aqueous solution	H319 eye irritation
Methanol	Dissolves the aldehyde	32.04	5–7 mL	Colorless liquid; bp 65 °C; d 0.79 g/mL; flash pt 11 °C	H225 highly flammable · H301+H311+H331 toxic · H370 organ damage (eyes/CNS)
Addition product (β-hydroxy ketone)	Product — Part 1 (RT)	272.26	— (MW only)	—	—
Condensation product (enone)	Product — Part 2 (50 °C)	254.24	— (MW only)	—	—

FYI › condensation MW = addition MW – 18 (loss of one H_2O). Each pair runs Part 1 *or* Part 2, then shares data.

GRAB FIRST: 100-mL round-bottom + stir bar · 1-mL syringe · cork ring (for weighing) · 25-mL Erlenmeyer · steam bath · funnel · stir plate on a raised lab jack (Part 2 also: heat block + condenser to 50 °C) · Büchner + filter flask + vacuum.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Reaction (Part 1 = RT · Part 2 = 50 °C)

1. Tare a cork ring; put a stir bar + 1-mL syringe in a **100-mL RBF**; weigh the assembly (record).
2. Draw **~0.2 mL 2-acetylpyridine** into the syringe at the dispensing station; reweigh assembly → **exact mass of 2-acetylpyridine**.
3. Add the 2-acetylpyridine into the RBF + **20 mL H_2O**. Rinse syringe w/ acetone into waste, discard.
4. In a 25-mL Erlenmeyer dissolve **300 mg 4-nitrobenzaldehyde in 5 mL MeOH** over a steam bath (add 2 mL more MeOH if needed). While warm, add it to the RBF via funnel.
5. Add **10 mL 0.54% Na_2CO_3** through the same funnel (base = starts the enolate).
6. Stir vigorously **60 min**. Part 1: room temp → stops at the addition product. Part 2: 50 °C (heat block + condenser, watch for flooding) → drives dehydration to the condensation product. Keep organic solvents away from the hot block.
7. Collect solid by **vacuum filtration**; wash **2 × 20 mL DI water**; pull vacuum 10 min; scrape onto filter paper & label (ADDITION or CONDENSATION).
8. Weigh → % yield; take mp / IR / NMR; compare with the other group's product.

✍ **WRITE**
limiting reagent (compare mol 2-acetylpyridine vs 4-nitrobenzaldehyde 300 mg ÷ 151.12) → theoretical product → % yield.

Experiment Notes (in lab)

Which part (RT / 50 °C): ________

Mass assembly: ________ g **after 2-acetylpyridine:** ________ g

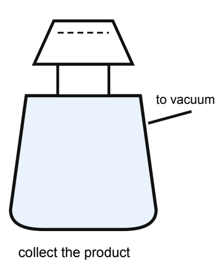
Mass 2-acetylpyridine: ________ g

Appearance during stir (solid forming? color):

Product mass / mp: ________

% yield · addition vs condensation compare:

Setup — draw this



vacuum filtration

CONTEXT (don't copy) ›

Addition vs condensation: the enolate adds to the aldehyde to give a β-hydroxy ketone (the *addition* product). Heating (and the electron-poor 4-nitro ring) makes it lose water (E1cb) to give the conjugated **α,β-unsaturated ketone** (the *condensation* product). Same starting materials — the temperature decides.

Why it precipitates: the conjugated nitro product is a bright, insoluble solid → easy vacuum filtration.

Clean-up & Conclusions

- Filtrates → aqueous waste; MeOH/organic → organic waste; syringe rinses → labeled waste; return glassware; bench task.
- **Conclusions** (~4 sentences after lab): which product your conditions gave and why (RT vs 50 °C); % yield; how mp/IR/NMR distinguish addition (has O–H, sp^3 C–H) from condensation (C=C, conjugation); comparison with the other group.